

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	SWTID-2026-7108
Project Name	BookStore
Maximum Marks	4 Marks

Technical Architecture

Note: The architectural diagram would be inserted here, illustrating a standard 3-tier MERN stack architecture mapping the user interface (React) to the backend server (Node.js/Express) and database (MongoDB) hosted on cloud infrastructure, similar to the reference models.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How user interacts with application (Web UI responsive perfectly on computers, tablets, and smartphones)	React.js, HTML, CSS, JavaScript
2	Application Logic-1	Logic for a process in the application (Book Search, Discovery, and Browsing operations)	Node.js, Express.js
3	Application Logic-2	Logic for a process in the application (Purchasing and Reservation checkout logic)	Node.js, Express.js
4	Application Logic-3	Logic for a process in the application (Generating personalized recommendations and managing reviews)	Node.js, Express.js

5	Database	Data Type, Configurations etc. for storing the vast collection of books across every genre, author, and language	MongoDB (NoSQL)
S.No	Component	Description	Technology
6	Cloud Database	Database Service on Cloud for highly available data storage	MongoDB Atlas
7	File Storage	File storage requirements for book cover images and platform assets	Cloud Object Storage (e.g., AWS S3)
8	External API-1	Purpose of External API used in the application (Payment Gateway for secure book purchasing)	Stripe API / PayPal API
9	External API-2	Purpose of External API used in the application (OAuth for user registration and authentication)	Google OAuth / LinkedIn API
10	Machine Learning Model	Purpose of Machine Learning Model (Engine to generate personalized recommendations based on user taste)	Collaborative Filtering Model (Python / Node.js)
11	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud to ensure users can explore anytime, anywhere	AWS / Heroku / Vercel

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	List the open-source frameworks used to build the platform	MERN Stack (MongoDB, Express.js, React, Node.js)
2	Security Implementations	List all the security / access controls implemented, use of firewalls etc., to ensure secure accounts and checkouts	JWT (JSON Web Tokens), bcrypt encryption, SSL/HTTPS
3	Scalable Architecture	Justify the scalability of architecture to handle a vast and growing book collection	3-Tier Architecture (React Frontend, Express API Backend, MongoDB)
4	Availability	Justify the availability of application so users can explore anytime, anywhere	Cloud deployment with load balancers and MongoDB Atlas high-availability clusters

5	Performance	Design consideration for the performance of the application to ensure a smooth, fast experience	React Virtual DOM, CDNs for static assets, Database Indexing in MongoDB
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Table-2: Application Characteristics

References

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- <https://c4model.com/>
 - <https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
 - <https://www.ibm.com/cloud/architecture>
 - <https://aws.amazon.com/architecture>
 - <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>
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